Lab 5 – Linear Antenna Arrays

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Prelab

(Obj-1)

A 10-element ordinary end-fire linear array is designed for operation at 1 GHz with an element spacing of λ/4 and the main beam directed along the positive z-axis. Uniform amplitude excitation with an appropriate progressive phase shift is used to satisfy the ordinary end-fire condition. The theoretical maximum directivity, HPBW, and SLL are calculated using array-factor theory.

Obj -1) $N = 10$ $d = \lambda/4$ $\lambda = \frac{c_0}{f} = 0.3\,m$ $\theta = 0$

$\beta = -kd = -\frac{\pi}{2}$

$$HPBW \simeq 2\cos^{-1}\left(1 - \frac{1.391\lambda}{\pi d N}\right) = 69.25^\circ$$

$SLL \simeq -13.46\,dB$

$$D_0 = \frac{4Nd}{\lambda} = 10\,dB$$

excitations

1	2	3	4	5	6	7	8	9	10
0	$-\frac{\pi}{2}$	$-\pi$	$-\frac{3\pi}{2}$	-2π	$-\frac{5\pi}{2}$	-3π	$-\frac{7\pi}{2}$	-4π	$-\frac{9\pi}{2}$

(Obj-2)

A 10-element Hansen–Woodyard end-fire array with element spacing λ/4 is designed following the conditions of Figure 6.14 in the textbook. Uniform amplitude excitation with Hansen–Woodyard phasing is applied to increase end-fire directivity. Theoretical values of maximum directivity and HPBW are computed using the array factor and compared to the ordinary end-fire case.

Obj-2) $N = 10$ $d = \lambda/4$ $\lambda = 0.3m$ $\theta = 0$

$$\beta \simeq -\left(kd + \frac{\pi}{N}\right) = -1.885\ rad$$

$$D_0 = \frac{7.214\,Nd}{\lambda} = 12.56\,dB \qquad SLL =$$

$$HPBW = 2\cos^{-1}\left(1 - 0.1398\frac{\lambda}{Nd}\right) = 38.5^\circ$$

excitations

1	2	3	4	5	6	7	8	9	10
0	-1.885	-3.77	-5.65	-7.54	-9.425	-11.31	-13.19	-15.08	-16.96

(Obj-3)

A 10-element broadside Dolph–Chebyshev array with λ/4 spacing and a specified sidelobe level of 35 dB is designed. Non-uniform amplitude excitation is determined, and the element spacing is verified to preserve the desired SLL. Theoretical maximum directivity and HPBW are calculated from the array factor.

Obj-3) $N=10$ $d=\lambda/4$ $\lambda=0.3\,m$ $R_0\,(dB)=35$ $M=\frac{N}{2}=5$

$$R_0 = 10^{R_0(dB)/20} = 56.23$$

$$z_0 = \cosh\left[\frac{1}{2M-1}\cosh^{-1}(R_0)\right] = 1.031$$

$$\frac{1}{2}(AF)_{2N} = \sum_{m=1}^{M} a_m T_{2m-1} = T_{2N-1}(z\cdot z_0) \Rightarrow \sum_{m=1}^{5} a_m T_{2m-1}(z) = T_9(z z_0)$$

From Slides:

$$256 a_5 = 256 z_0^9$$

$$64a_4 - 576a_5 = -576 z_0^7$$

$$16a_3 - 112a_4 + 432a_5 = 432 z_0^5$$

$$4a_2 - 20a_3 + 56a_4 - 120a_5 = -120 z_0^3$$

$$a_1 - 3a_2 + 5a_3 - 7a_4 + 9a_5 = 9z_0$$

$$A = [\overset{a_1}{18.6},\ 15.96,\ 11.57,\ 6.83,\ \overset{a_5}{3.27}]$$

$$\beta = 0$$

$$d_{Max} \le \frac{\lambda}{\pi}\cos^{-1}\left(-\frac{1}{z_0}\right) = 0.2765$$

$$d = 0.075 < d_{Max} \checkmark$$

$$D_0 \approx \frac{2R_0^2}{1+(R_0^2-1)\frac{\delta\cdot\lambda}{L+d}} \qquad \delta = 1 + 0.636\left[\frac{2}{R_0}\cosh\sqrt{(\cosh^{-1}R_0)^2-\pi^2}\right]^2$$

$$HPBW = \delta\left|\cos^{-1}\left(-\frac{2.787}{Nkd}\right) - \cos^{-1}\left(\frac{2.787}{Nkd}\right)\right|$$

$D_0 = 6.49\,dB$

$HPBW = 25.15°$

$SLL = 35\,dB$ by design

(Obj-4)

The excitation coefficients obtained in Objectives 1–3 are used to generate custom array definition text files for each array configuration, ensuring correct syntax and clear identification of each array type.

Lab

Part 1 — Resonant Half-Wave Dipole

Figure 1 shows the model setup for the single resonant linear half-wave dipole antenna. The dipole is constructed from copper wire aligned along the x-axis with a small central gap. The structure is enclosed in an air box with radiation boundary conditions applied.

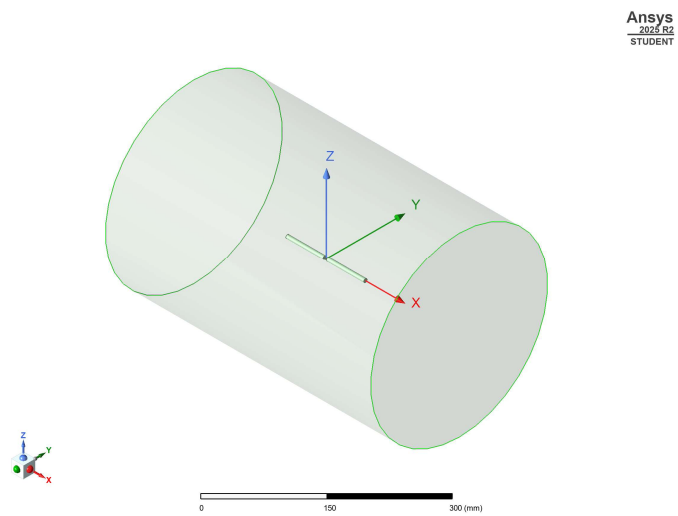

Figure 1: Model setup for Part 1 resonant half-wave dipole antenna.

Figure 2 shows the port configuration for Part 1. A Driven Modal lumped port with a 75 Ω termination is applied across the dipole gap to excite the antenna.

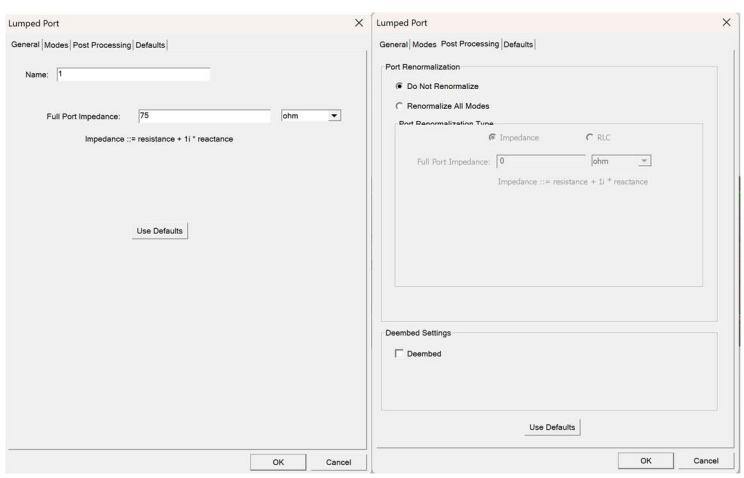

Figure 2: Lumped port configuration for resonant dipole antenna (Part 1).

(Res 1) The resonant dipole length was found to be 66 mm, yielding an input impedance of 71.36 − j0.07 Ω at 1 GHz. This confirms near-resonant behavior with minimal reactive component.

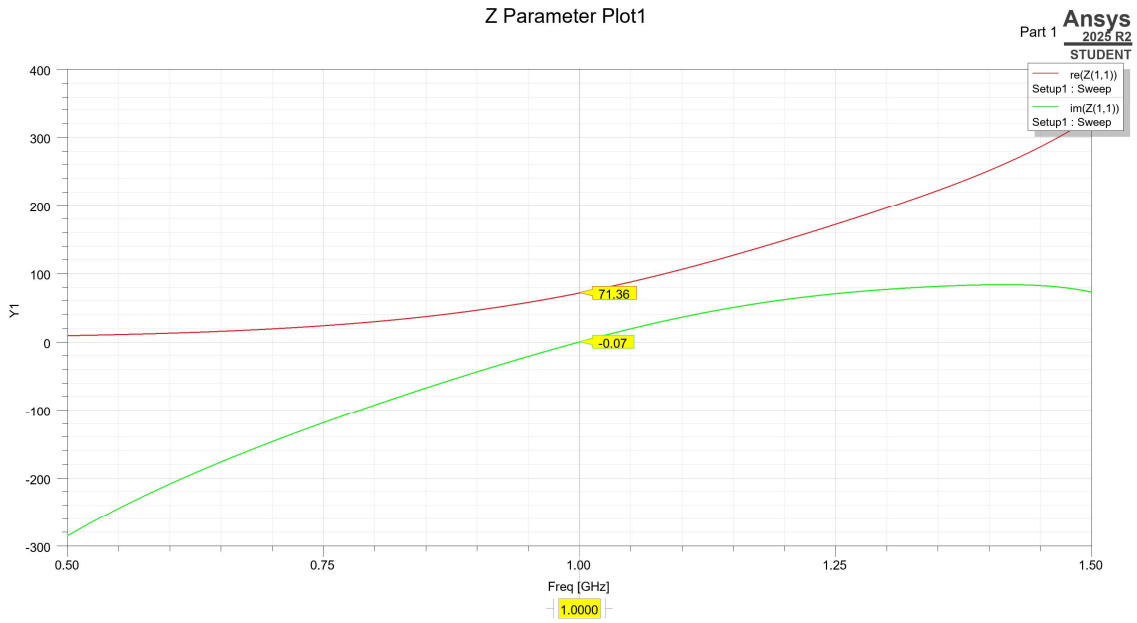

Figure 3: Input impedance of resonant half-wave dipole versus frequency

(Res 2) The S-parameter plot confirms resonance near the design frequency.

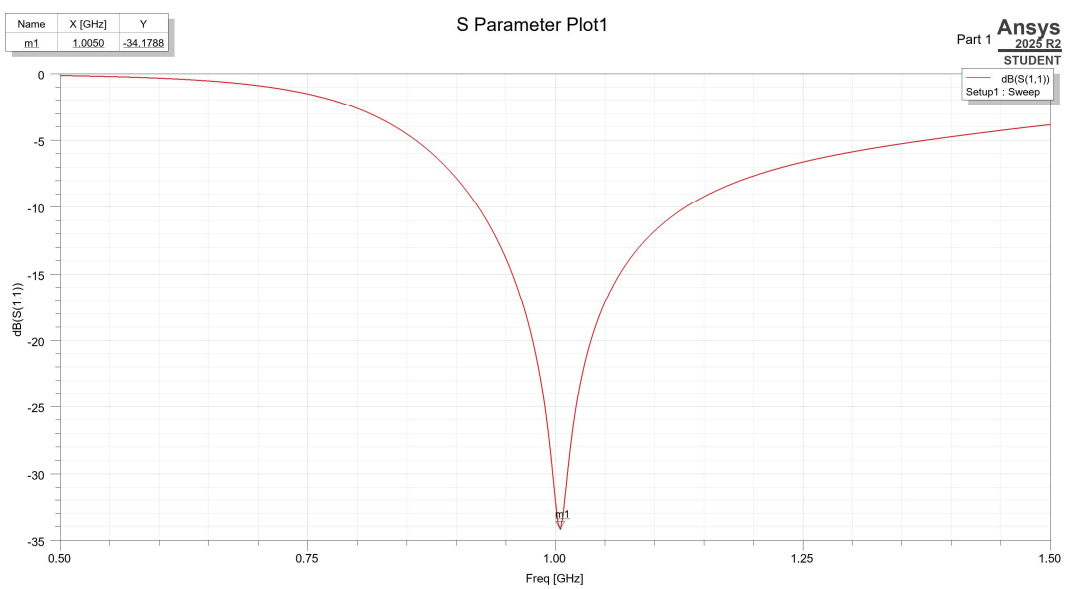

Figure 4: S(1,1) plot in dB for resonant half-wave dipole versus frequency

(Res 3) As seen in Figure 5, the gain plot matches the expectation for a dipole antenna oriented along the x-axis.

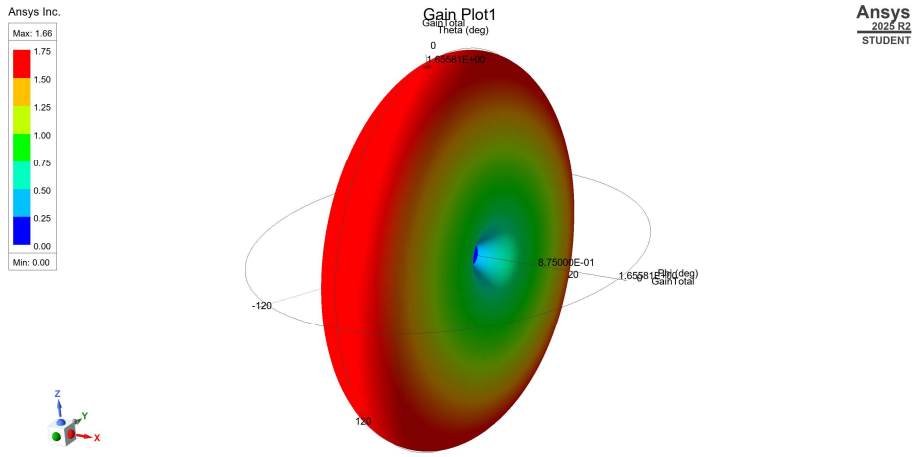

Figure 5: 3D Polar Gain plot for resonant half-wave dipole

Part 2 — Ordinary End-Fire Array

For **Res-3 and Res-4**, the simulation was modified to explicitly model the full array. The antenna element geometry and lumped port definitions remain unchanged from the single-element case, but the array is constructed using physically replicated elements rather than a custom array excitation, and the air box is adjusted to match. This change is required because custom array files do not support analysis of S- and Z-parameters.

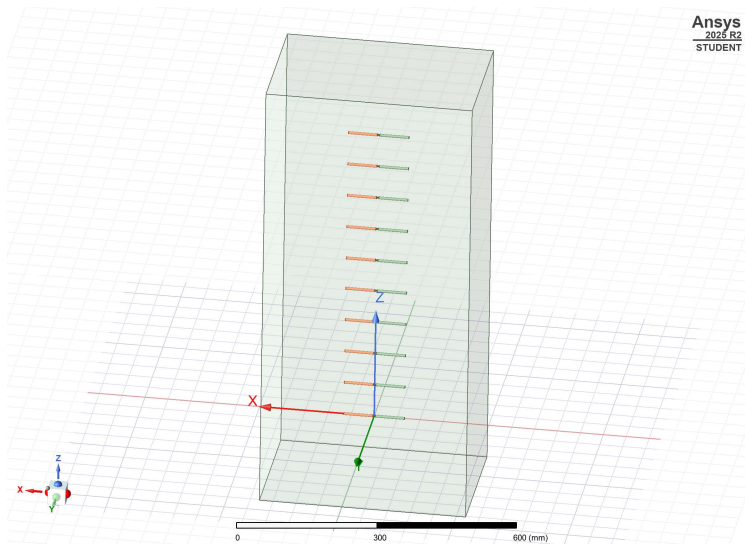

Figure 6: Modified array model used for S- and Z-parameter analysis.

(Res 4) Figure 7 shows the scattering parameters of the ordinary end-fire array. S(5,5) indicates element 5 is best matched near 1 GHz. According to S(5,1) and S(5,6), coupling is strongest between neighboring elements and much weaker between distant elements. Furthermore, coupling effects are stronger near the design frequency.

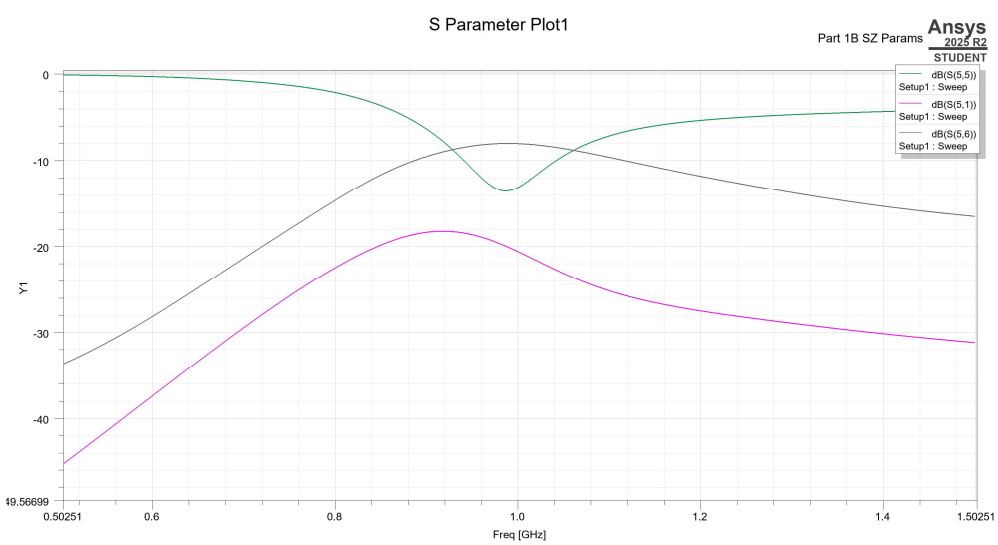

Figure 7: Scattering parameters of ordinary end-fire array

(Res 5) Figure 8 shows the impedance parameters of array elements as a function of frequency. The impedance of element 5 is closest to purely real near 1 GHz, indicating that the element is well tuned at the design frequency when operating within the array. Away from this frequency, the impedance becomes more reactive, showing that array interactions affect element behavior.

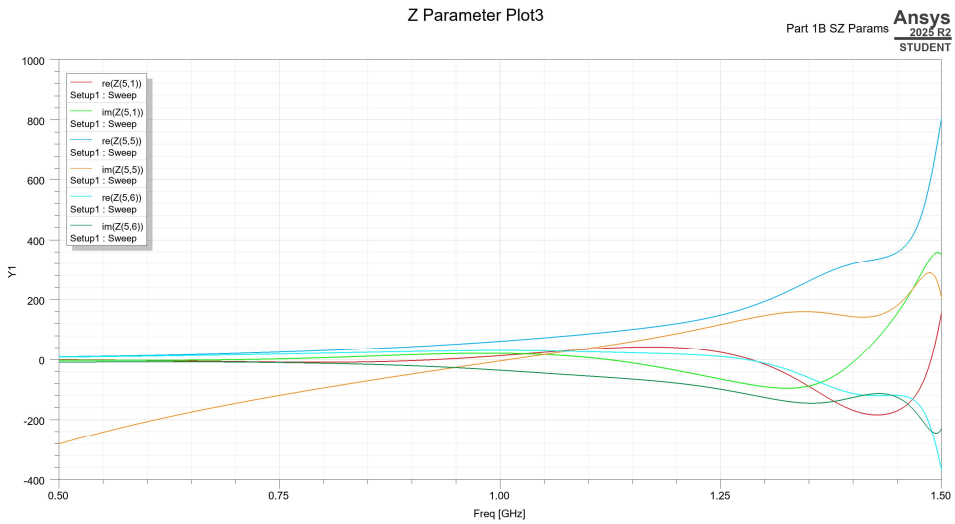

Figure 8: Impedance parameters of the ordinary end-fire array showing real and imaginary components versus frequency.

For Res-6 and Res-7, the antenna element geometry and port setup from Part 1 are reused without modification. Element excitations are defined using a custom array file implementing ordinary end-fire phasing. Figure 9 illustrates the linear array configuration formed by applying the Custom Array file excitations.

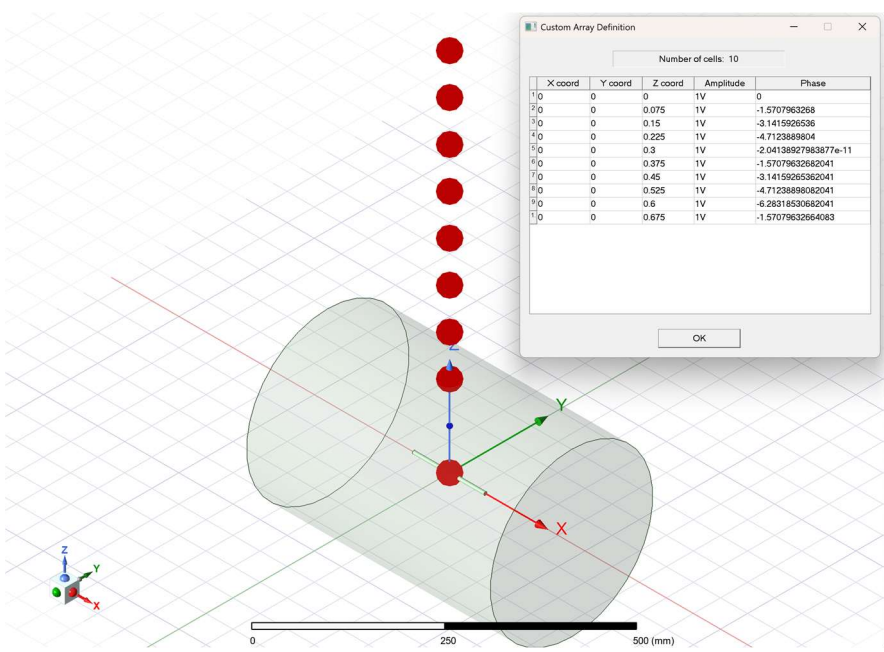

Figure 9: Linear ordinary end-fire array geometry and excitations used in Part 2.

(Res-6) Figure 10 shows the three-dimensional gain of the ordinary end-fire array, clearly exhibiting end-fire radiation along the array axis, consistent with the applied progressive phase excitation.

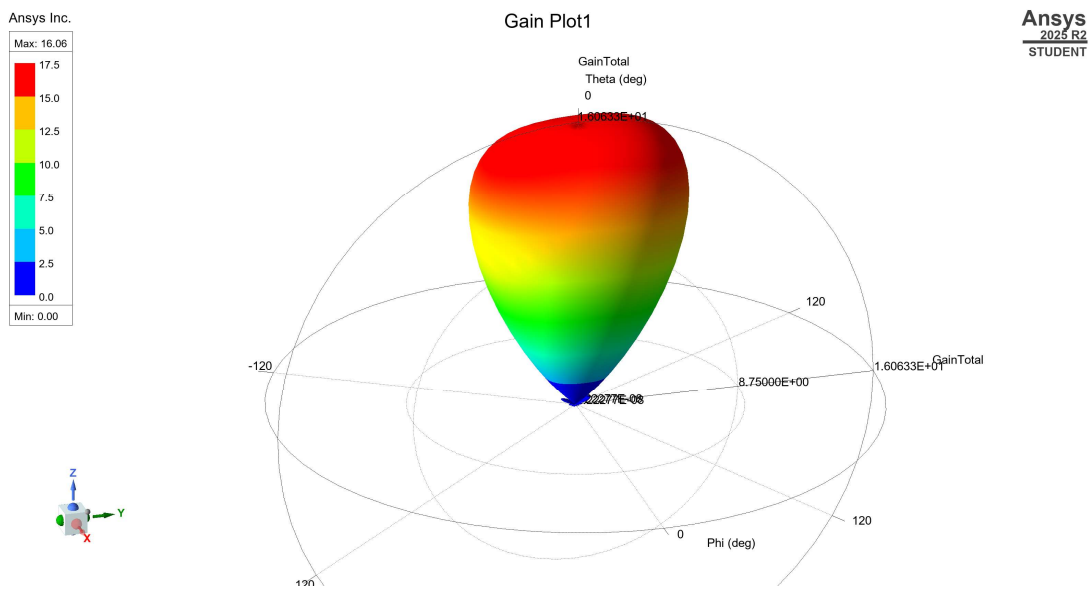

Figure 10: 3D gain pattern of the ordinary end-fire array at 1 GHz.

(Res-7) Figure 11 plots the θ-domain directivity in dB for the Part 2 array with φ held constant at π/2. Markers indicate the simulated maximum directivity and half-power beamwidth, which are compared to theoretical values. The plot closely matches Figure 6.9 in the textbook.

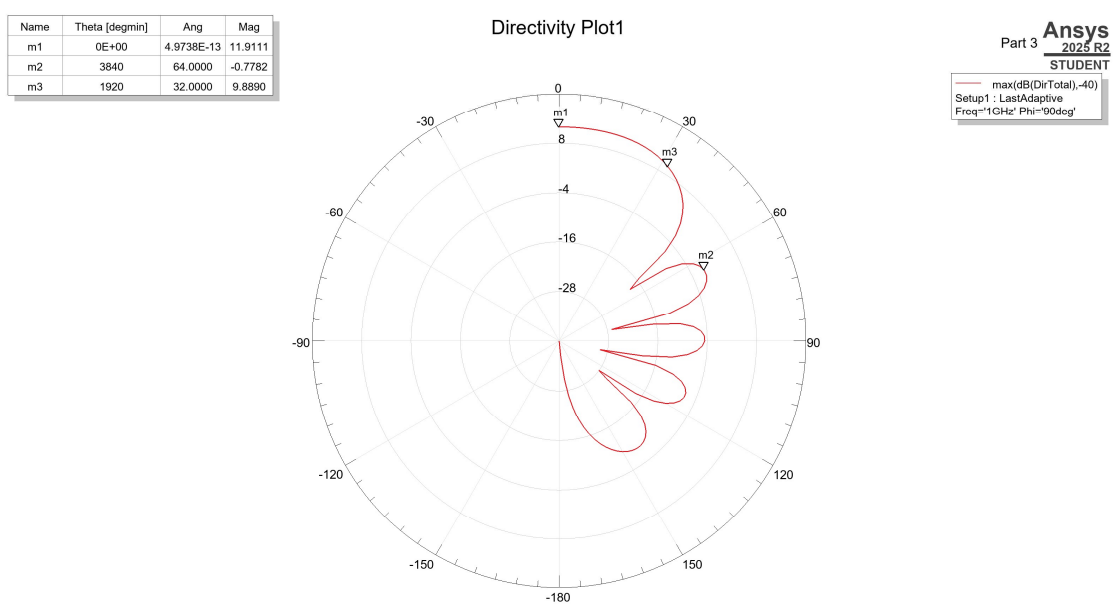

Figure 11: 2D directivity of the ordinary end-fire array (φ = π/2).

(Res 8) The ordinary end-fire array exhibits a simulated HPBW of 64° compared to the theoretical 69°. The sidelobe level was −12.7 dB versus the theoretical −13.46 dB, and the maximum directivity was 11.91 dB compared to the theoretical 10 dB. Overall agreement is strong, with discrepancies attributed to finite array size and element pattern effects.

Part 3 — Hansen–Woodyard End-Fire Array

The Hansen–Woodyard array reuses the exact antenna geometry, air box, and port configuration from Part 2. Only the custom array definition file is changed to apply Hansen–Woodyard phasing. Figure 12 shows the corresponding custom array configuration.

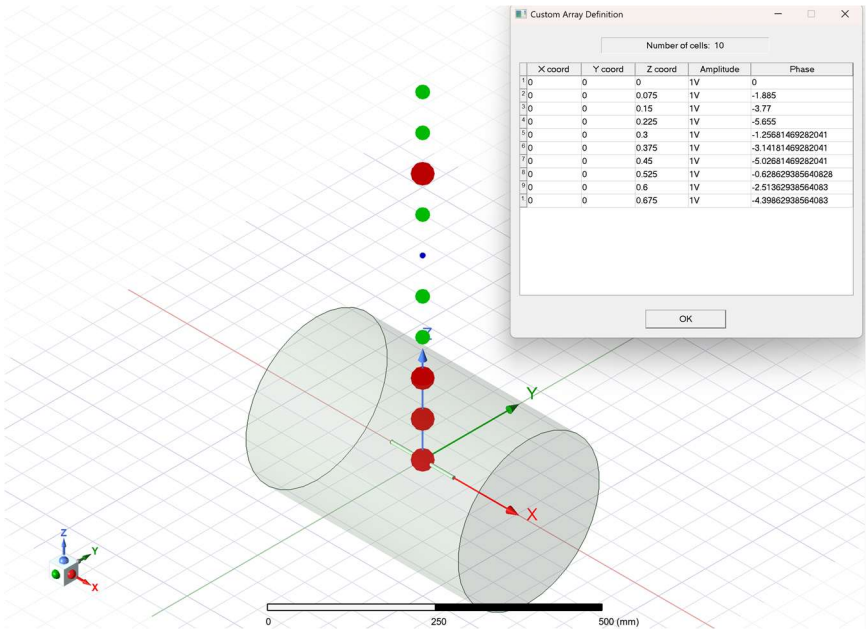

Figure 12: Custom array definition for Hansen–Woodyard end-fire array.

(Res-9) Figure 13 shows the θ-domain directivity of the Part 3 Hansen–Woodyard array with φ = π/2. Markers identify the simulated maximum directivity, half-power beamwidth, and sidelobe level, which are evaluated against theoretical expectations. The plot closely matches Figure 6.14 in the textbook.

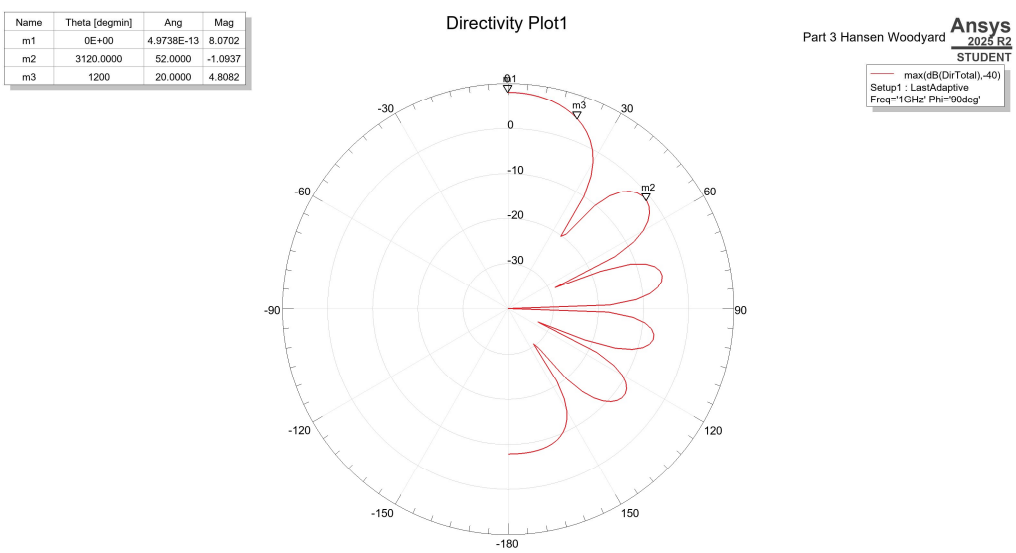

Figure 13: 2D directivity of the Hansen–Woodyard end-fire array (φ = π/2).

(Res 10) The Hansen–Woodyard array shows a narrower main beam (HPBW = 40° vs. 38.5° theoretical) but reduced maximum directivity (8.07 dB vs. 12.56 dB theoretical). This deviation is expected for small N, where Hansen–Woodyard phasing does not fully realize its asymptotic advantage.

(Res-11) Compared to the ordinary end-fire array from Part 2, the Hansen–Woodyard array exhibits a narrower main beam but reduced maximum directivity, as increased sidelobes draw power away from the main lobe for the small array size (N = 10).

Part 4 — Dolph–Chebyshev Broadside Array

The Dolph– Chebyshev broadside array also uses the same physical model and port setup as Part 2. A new custom array definition file is loaded to apply the Dolph– Chebyshev amplitude taper. Figure 14 shows the custom array configuration used for this case.

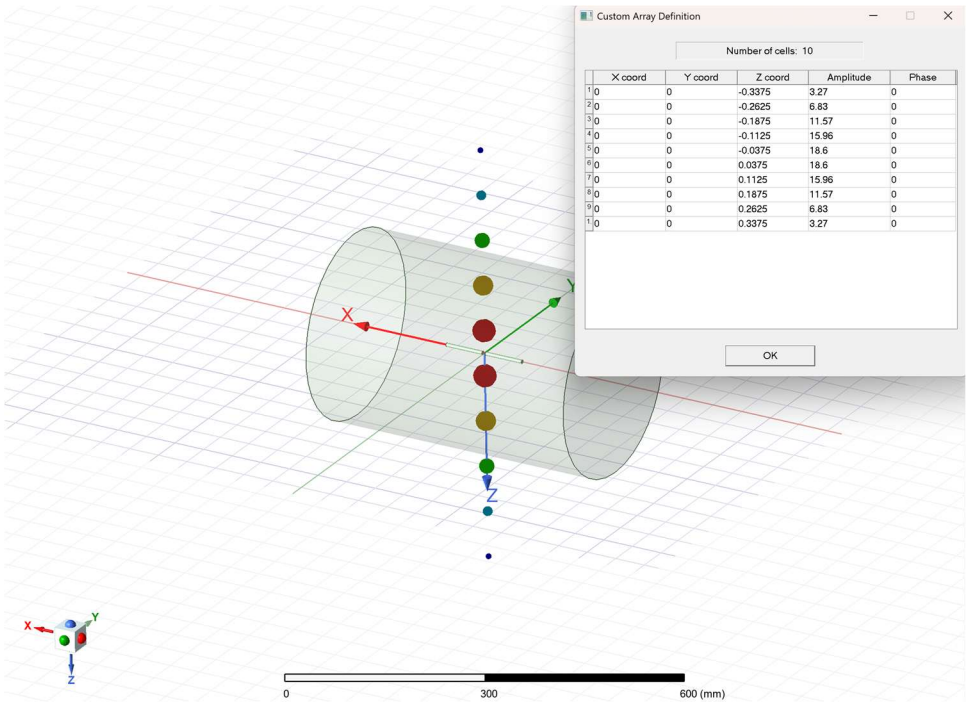

Figure 14: Custom array definition for Dolph– Chebyshev broadside array.

(Res-12) Figure 15 presents the θ-domain directivity of the Part 4 Dolph– Chebyshev array with φ held at π/2. Markers indicate the simulated maximum directivity, half-power beamwidth, and sidelobe level, demonstrating the intended sidelobe suppression.

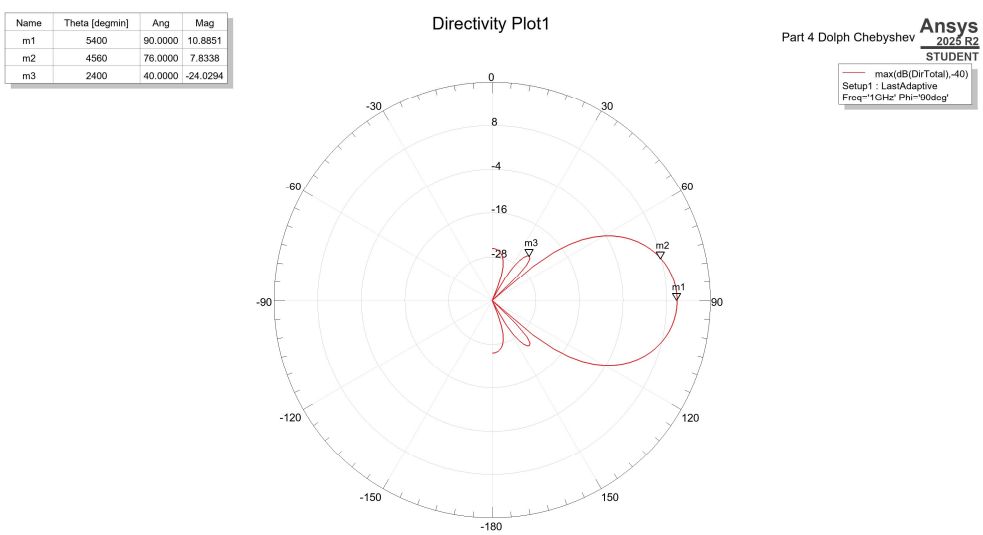

Figure 15: 2D directivity of the Dolph–Tschebyscheff broadside array ($\varphi = \pi/2$).

(Res 13) The Dolph–Tschebyscheff broadside array achieves strong sidelobe suppression (−34.9 dB vs. −35 dB theoretical) and an HPBW of 28° compared to the theoretical 25.15°. The simulated directivity exceeds the theoretical value, likely due to element pattern contributions.

Report Checklist

1. I have included screenshots of all the HFSS models. **AM**
2. I have included screenshots of all port setups, including integration lines where appropriate. **AM**
3. I have included all necessary results in graphs, tables, or statements. **AM**
4. I have interpreted and commented on all my results. **AM**